Gaurav Mahale

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Education

University of Glasgow Glasgow, Scotland
MSc., Data Science.
Relevant Coursework: Web Sciences, Text as Data, Deep Learning, Big Data, Information Retrieval

Pimpri Chinchwad College of Engineering Pune, India
B.Tech, Computer Science.
Relevant Coursework: Software Project Management, Database Management System, Machine Learning

Projects

App Based Digital Audiometer

- Developed a Flutter-based digital audiometer application as an affordable alternative to conventional hearing test systems.
- Implemented calibration, background noise monitoring, and full-range hearing tests, with automatic audiogram generation.
- Maintained a centralized database to support system updates and large-scale hearing assessments.
- Added multilingual support, chatbot assistance, and doctor referral features to help identify hearing loss and guide users towards treatment.
  Technologies: Flutter, Dart, Database Management, Mobile Application Development

Brain Tumour Detection using ML

- Built a brain tumour detection system using machine learning and image processing techniques.
- Trained and tested models to detect and classify tumours from MRI scan images.
- Automated image pre-processing and analysis to support early-stage diagnosis.
- Focused on improving accuracy and reducing manual review effort in medical imaging.
  Technologies: Python, Machine Learning, Image Processing, OpenCV, NumPy, Scikit-learn

Leadership & Activities

Member, Coding Club University of Glasgow

- Participated in coding workshops, collaborative projects, and peer-learning sessions to enhance programming skills

Treasurer, Desi Society University of Glasgow

- Managed society finances, prepared budgets, tracked expenses, and ensured smooth allocation of funds for events and activities.

Skills & Interests

Technical: C, C++, MySQL (stored procedures, views, triggers), MS Office (Word, Excel, PowerPoint), Power BI, Figma, Flutter, Python (NumPy, OpenCV, scikit-learn), Java, Git/GitHub.
Soft Skills: Problem Solving, Critical Thinking, Communication, Adaptability, Attention to Detail, Leadership and Teamwork, Event Coordination.
Language: English – Fluent, Hindi – Native.
Laboratory / Tools: Digital Audio Processing tools, Image Processing tools.

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Additional Projects

ReliableRAG - Production RAG with Hallucination Evaluation

- Combined BM25 and dense LSA retrieval with reranking to find relevant evidence.
- Added source citations and abstention when retrieved evidence was insufficient.
- Built an evaluation workflow covering retrieval, faithfulness, citation correctness, and abstention.
- Packaged the application with FastAPI, Streamlit, Docker, and automated checks.

Technologies: Python, FastAPI, Streamlit, BM25, LSA, Docker

Bank Marketing Prediction - Leakage-aware machine learning

- Built preprocessing and model-training pipelines for predicting term-deposit subscription.
- Excluded call duration from the deployable feature set to avoid information leakage.
- Compared baseline, logistic regression, decision tree, and random forest models.
- Added classification evaluation, feature importance, and a Streamlit prediction interface.

Technologies: Python, Pandas, Scikit-learn, Streamlit